

**Detroit Transportation Survey: Multi-Stage Area Probability Sample Design: A Technical
Sampling Report**

Effect Size

University of Maryland

Author Note

Correspondence concerning this article should be addressed to Effect Size, Email:
email.umd.edu

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Introduction

This report describes the design of a multi-stage area probability sample for a face-to-face survey of Detroit, Michigan residents. The survey examines transportation security and transportation use, with key outcome variables including a) the proportion of residents in households without a vehicle, b) the proportion whose main transportation is public transit, c) the proportion who skip destinations due to transportation problems, and d) the proportion who sought transportation help from friends, family, or neighbors.

The sample uses a two-stage selection of census tracts (PSUs) and block groups (SSUs). Since there is no address listing or roster of persons exists prior to fieldwork, we do not execute person-level selection in this report. Instead, we describe how field teams would list households in each selected block group and select respondents through those households, which is a standard approach in area probability sampling when absent person-level frame (VDK Ch. 10; Lecture Ch. 10 Slide 21). The key outcome variables are household-level (e.g., vehicle ownership is shared among all members of a household), which further motivates reaching persons through households and selecting only one adult per household to avoid correlated observations.

The design goals are to yield 1,000 total respondents while achieving a self-weighting sample within each age group and equal interviewer workload in each sample PSU. These dual objectives are accomplished through the composite measure of size (MOS) technique described in VDK Chapter 10, Section 10.5. This report will cover 1) the construction and cleaning of the sampling frame, 2) composite MOS computation and verification, 3) sample selection via systematic PPS, 4) anticipated precision, 5) a description of the weighting plan, 6) variance estimation guidance, 7) maps of the frame and sample, and 8) a sample listing appendix.